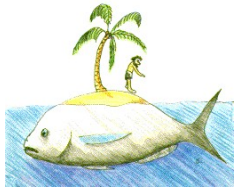


Can we model the spatial distribution of ecological groups during the last two decades based on spatial environmental drivers?

Marcela C. Nascimento; Torstein Pedersen; Berengérè Husson and Lillia Guillet

Main questions



Q1 How well can a spatial dynamic model predict spatial distributions of functional groups in the last two decades?

Q2 Can the model predict the changes in observed spatial distribution of the functional groups from cold to warm years?

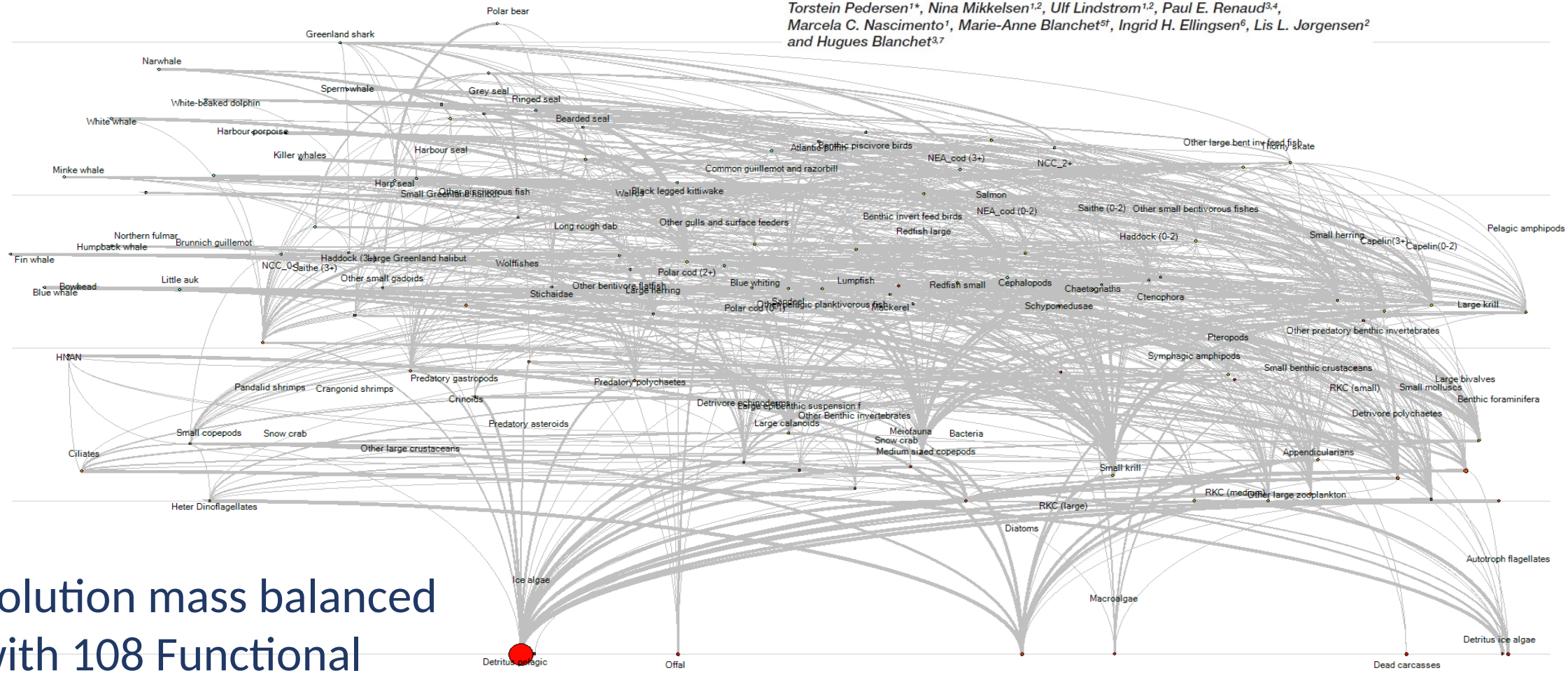
Trophic flow

frontiers in Marine Science

ORIGINAL RESEARCH
published: 17 September 2021
doi: 10.3389/fmars.2021.732637

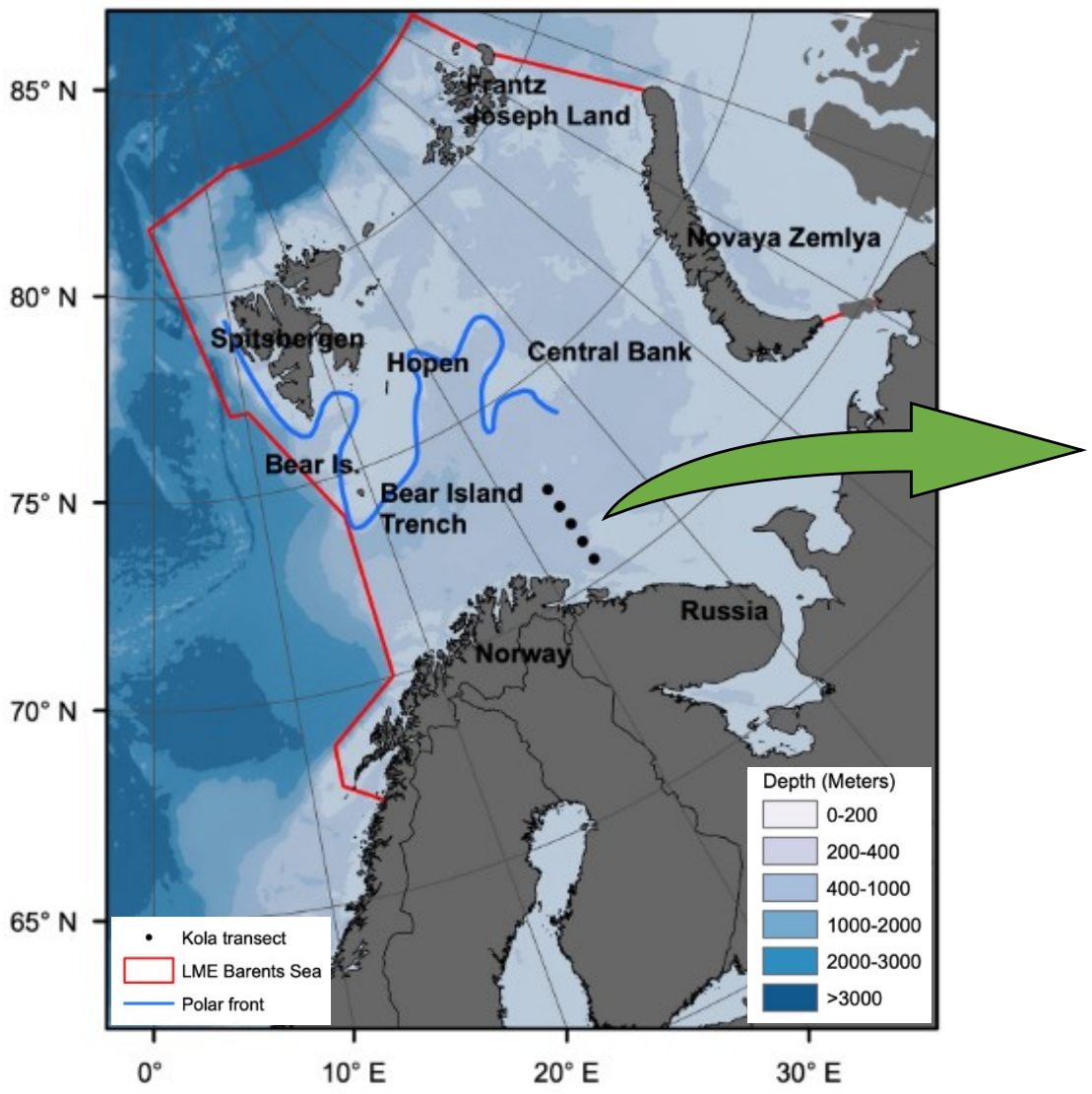
Overexploitation, Recovery, and Warming of the Barents Sea Ecosystem During 1950–2013

Torstein Pedersen[1], Nina Mikkelsen[1,2], Ulf Lindstrøm[1,2], Paul E. Renaud[3,4], Marcela C. Nascimento[1], Marie-Anne Blanchet[5†], Ingrid H. Ellingsen[6], Lis L. Jørgensen[2] and Hugues Blanchet[3,7]*

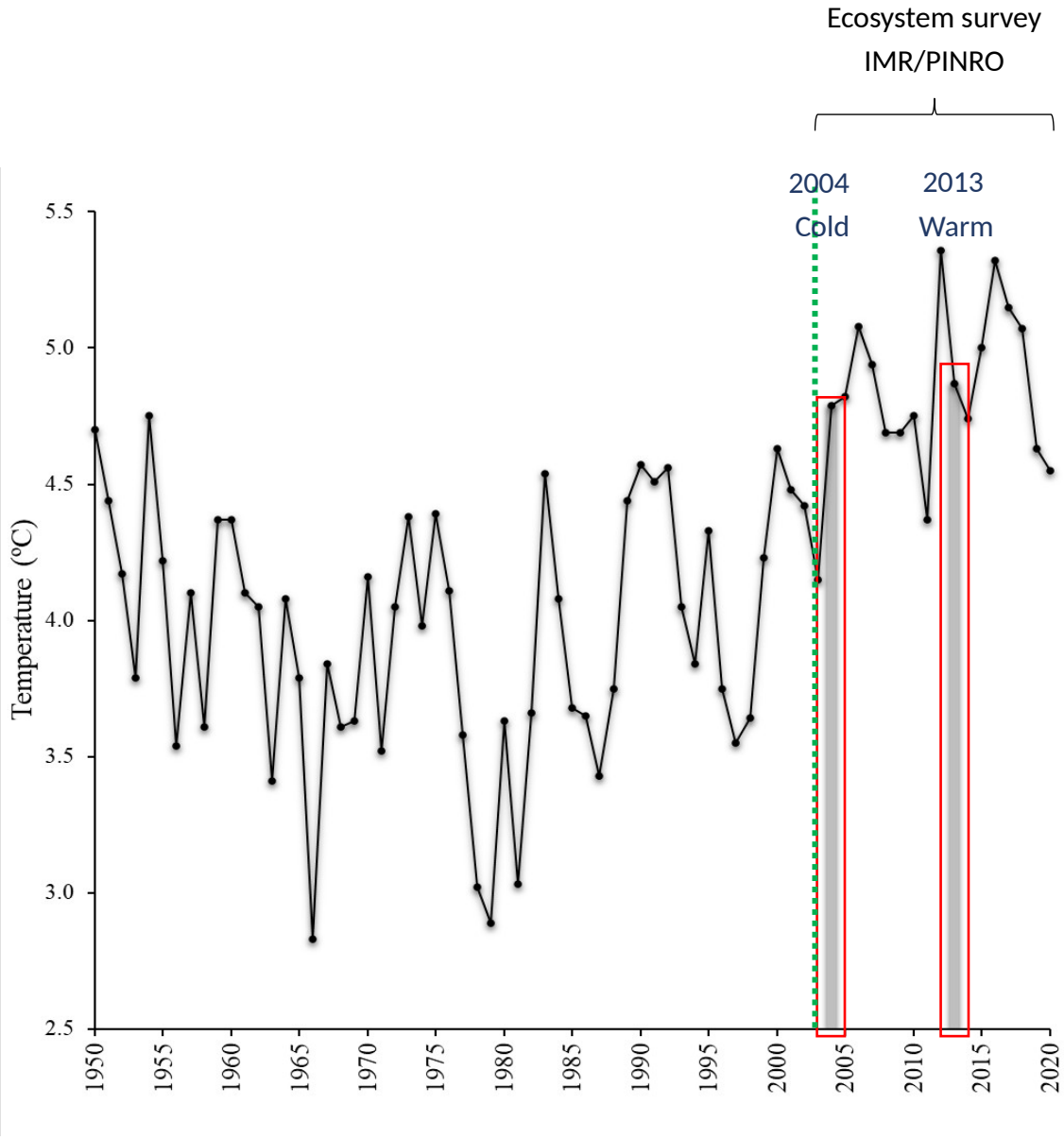


High resolution mass balanced model with 108 Functional groups (FG)

Environment the Barents Sea LME

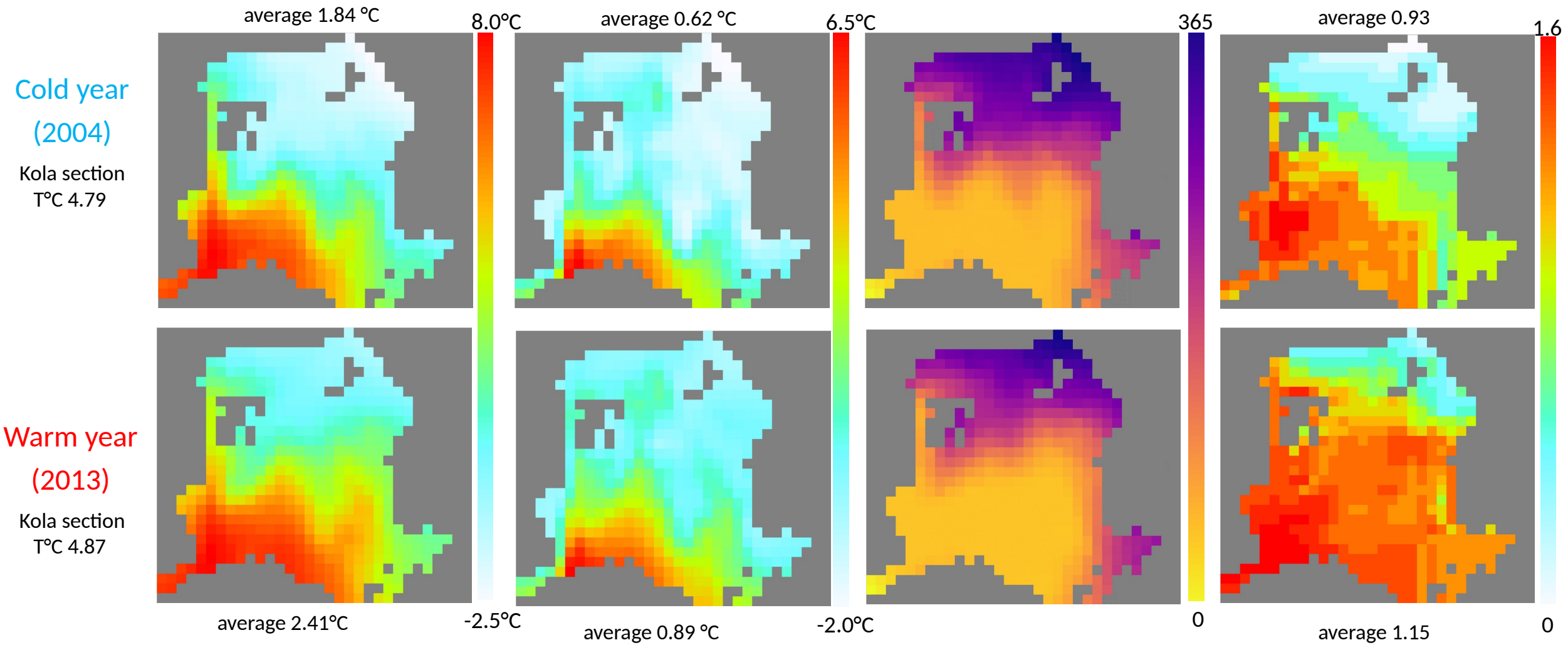


Pedersen et al (2021)



Kola section temperature 0-200 m depth

Basemaps used in scenarios

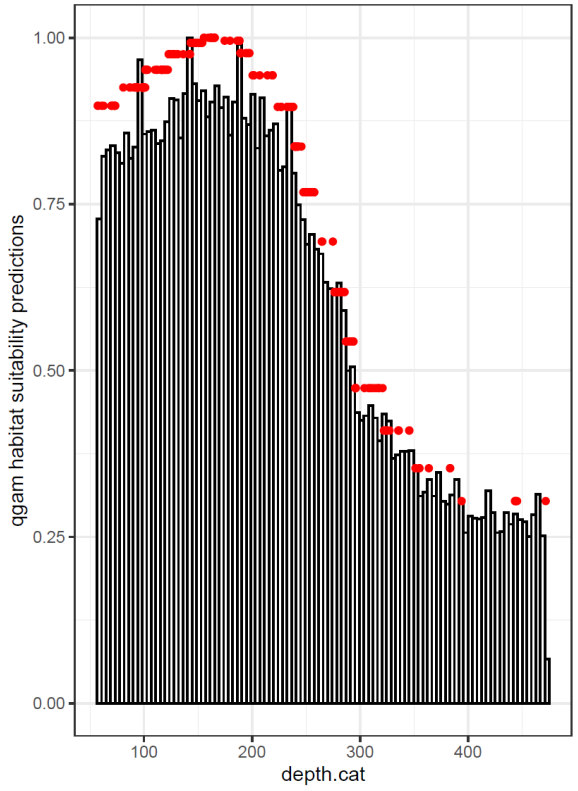


Environmental condition - SVIM archives

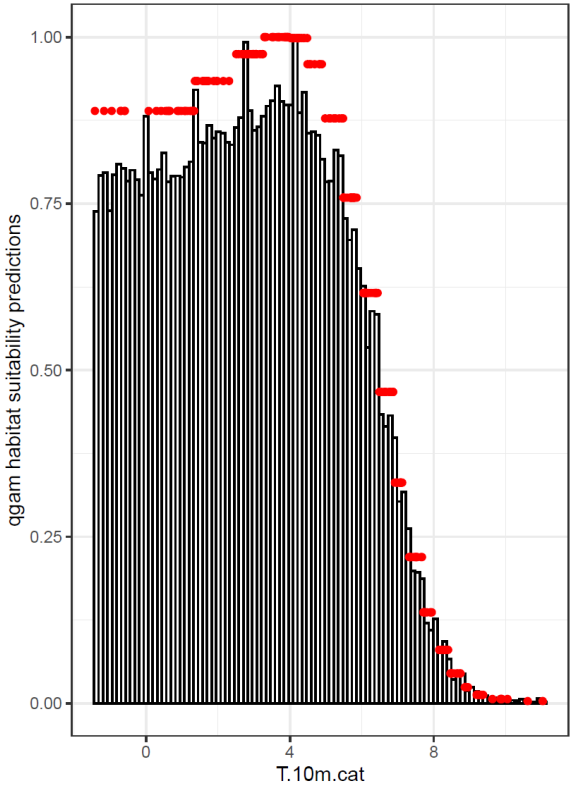
Primary production - Reigstad et al. (2011)

Habitat capacities

QGAM model – ecosystem survey/literature



Depth



Surface temperature

Thysanoessa (small krill)



http://www.arcodiv.org

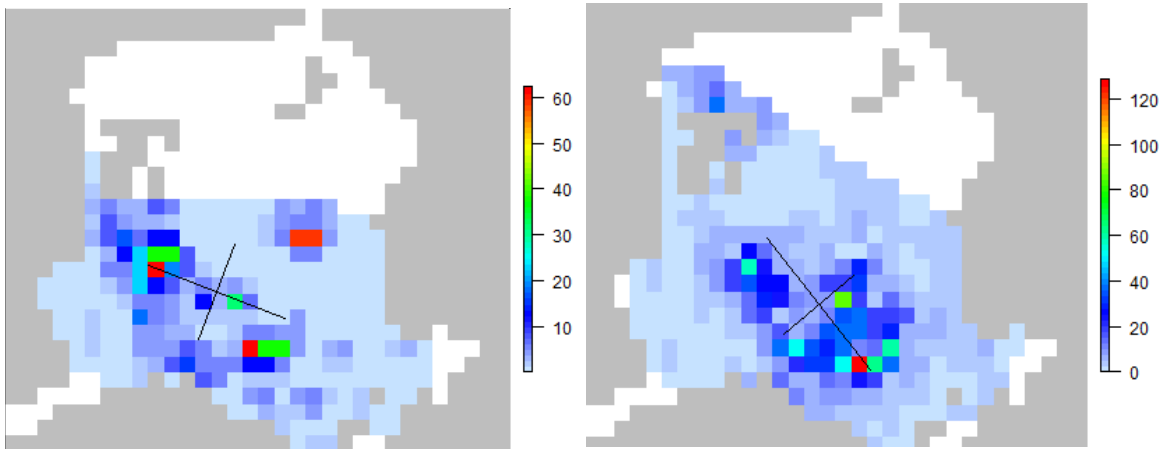
74 FG
1 or + environmental response

Functional groups with high biomass and good distribution information
35 FGs

Biomass distribution observed vs modelled

Observed

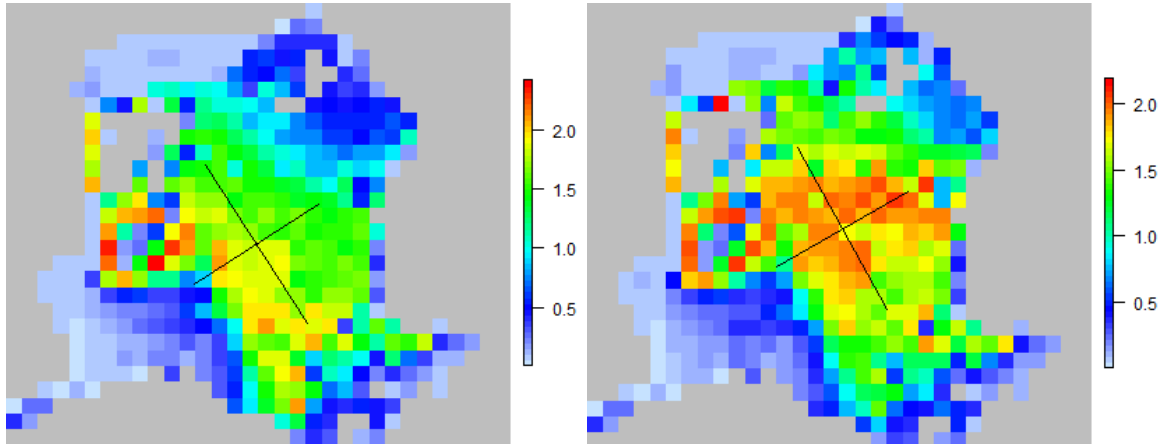
Cold year



Warm year

Mikhina et al. (2019), Eriksen and Dalpadado (2011) and Orlova et al. (2010)

Modelled

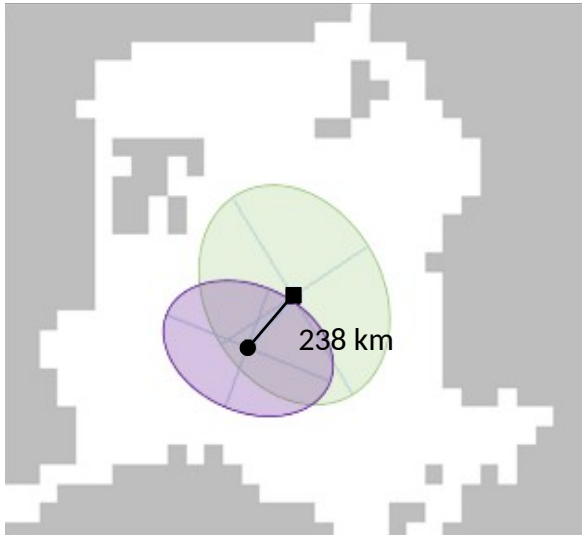


Thysanoessa (small krill)

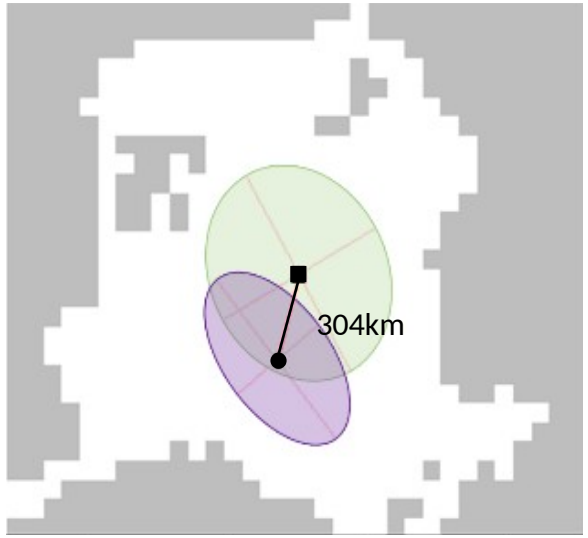


http://www.arcodiv.org

Cold year



Warm year

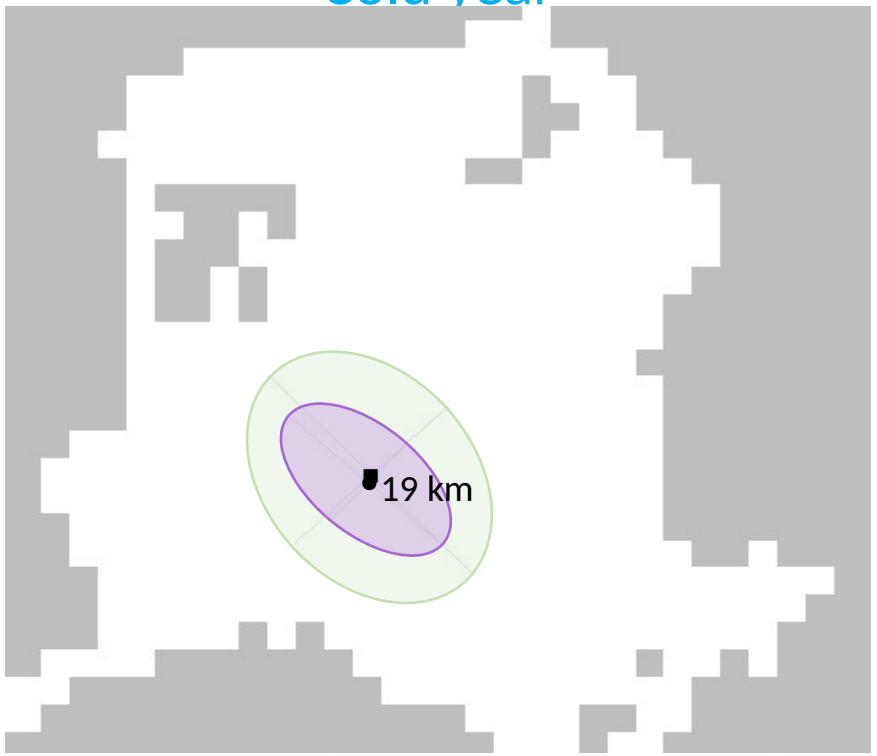


Modelled ▪ Observed •

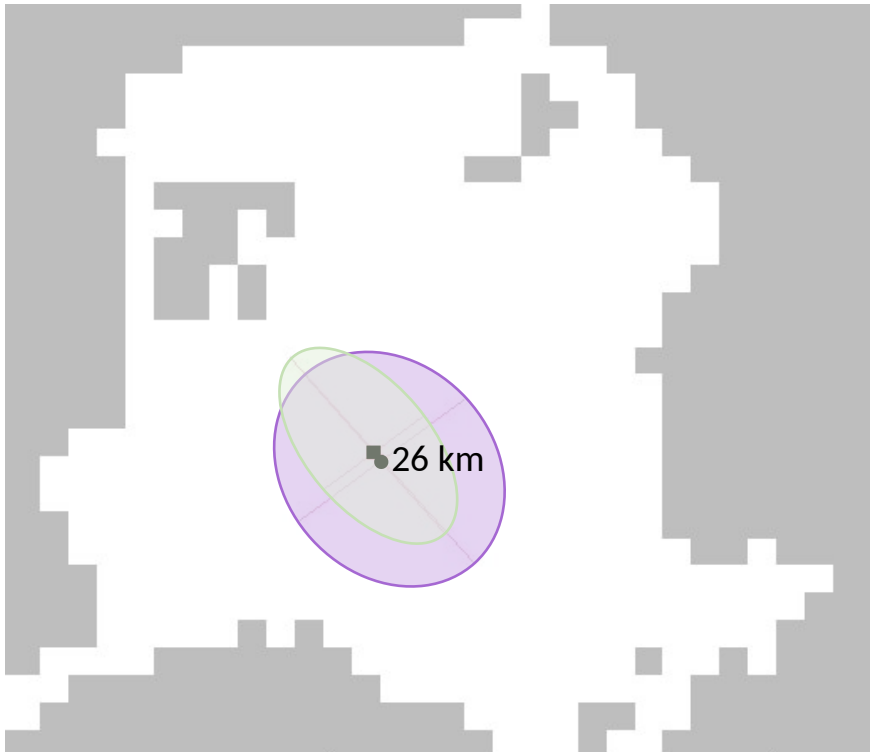
Biomass distribution observed vs modelled

Average directed difference between CG position of modelled and observed of the 35 FGs

Cold year



Warm year



Modelled ▪ Observed •

Shift in distributions from cold to warm year

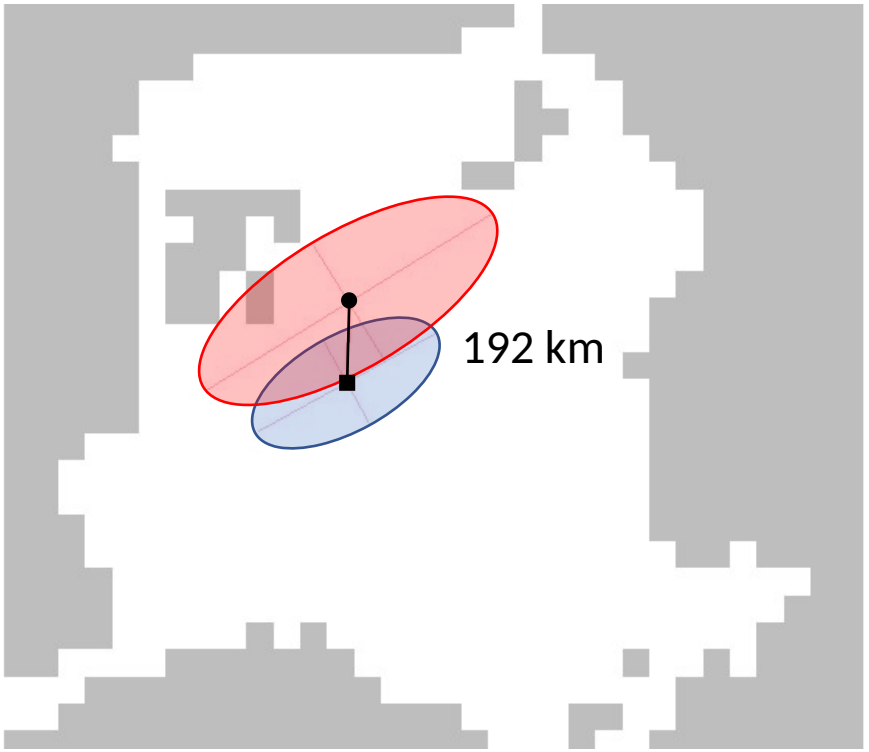
35 FGs

Capelin (3+)

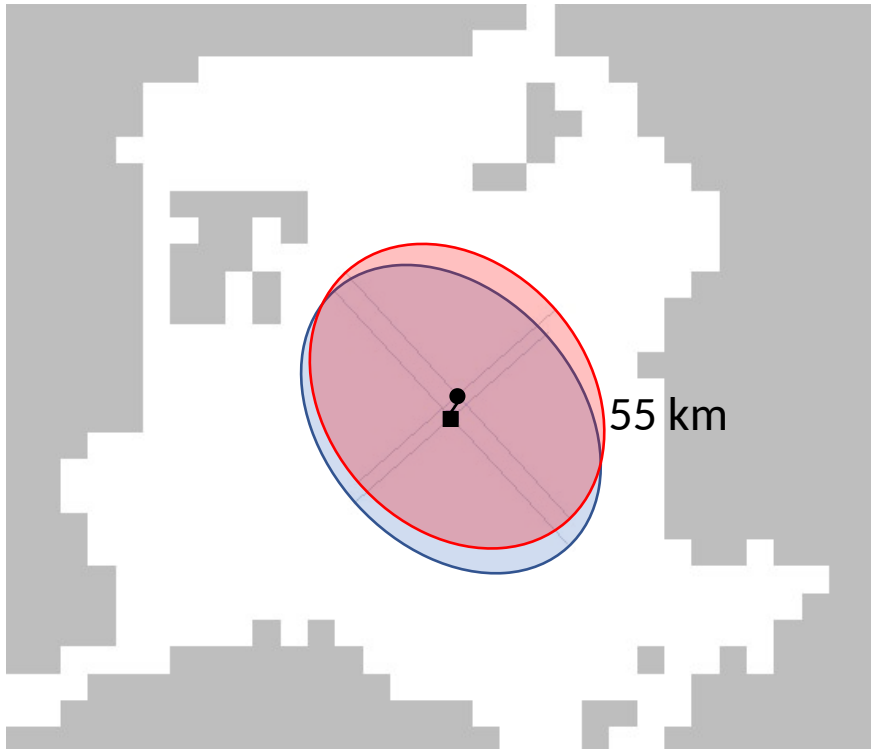


Danita Delimont (www.imago-images.com)

Observed

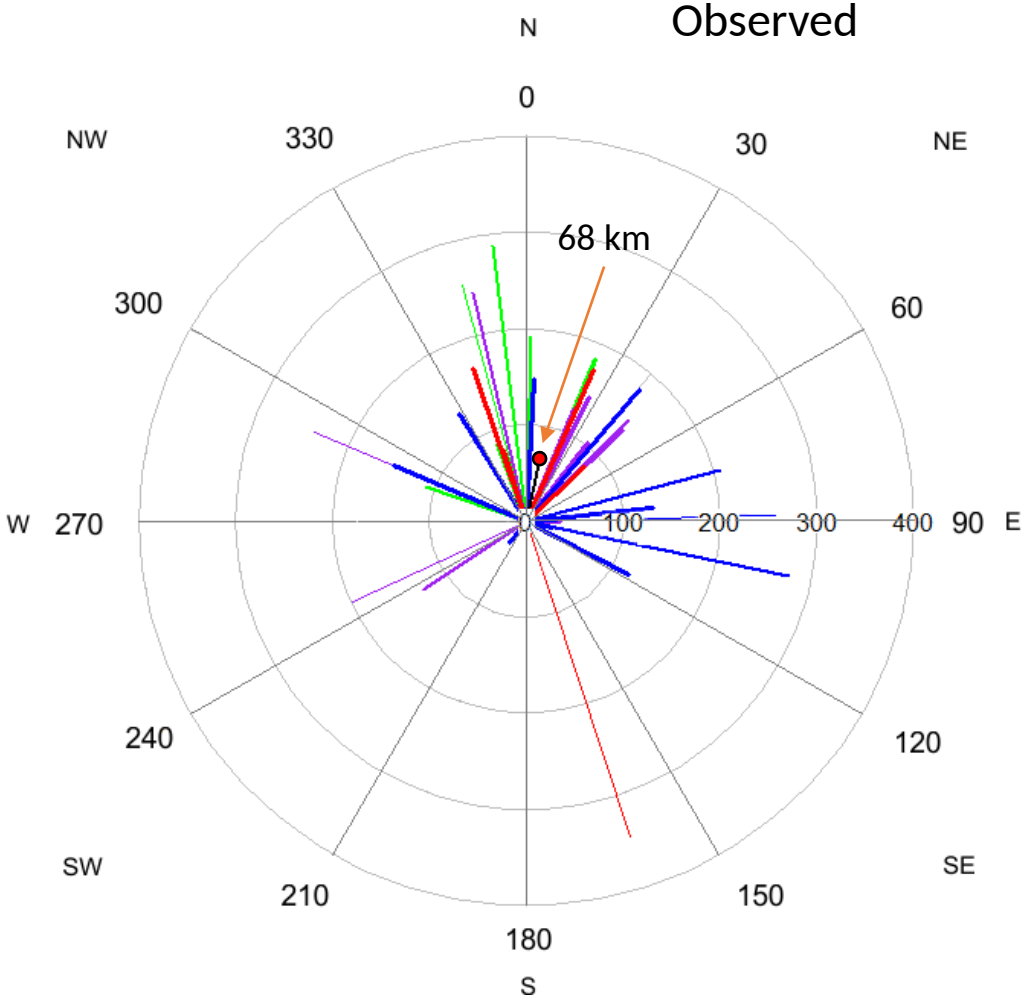


Modelled



Shift in distributions from cold to warm year for 35 FG

Observed

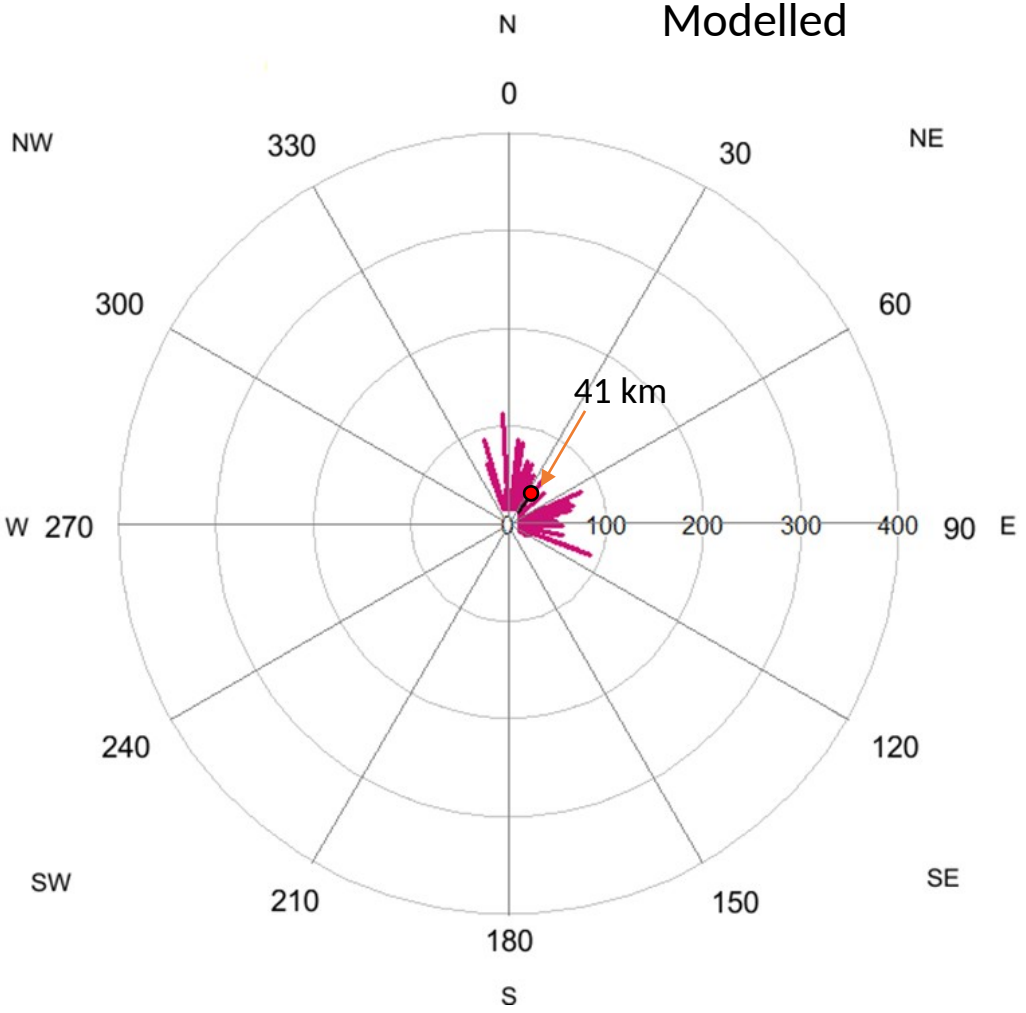


Thickness = **Overlapping (GIC)**
0 -1
Thinner - thicker

Colors = **uncertainty score**
(lower –higher)

Green = 1 (lower)
Blue = 2
Purple = 3
Red = 4 (higher)

Modelled



Average directed shift in CG position

Summarizing

Q1 How well can a spatial dynamic model predict spatial distributions of functional groups in the last two decades?

The model can represent well the FGs distributions.

Q2 Can the model predict the changes in observed spatial distribution of the functional groups from cold to warm years?

In general, the model can predict well the direction of shift, mainly north-eastern, considering the average position of the CG the direction and the distance of the model shift are very similar to the survey based observed shift.

Thank you



Common Gull *Larus canus*
FG 21 (Other gulls and surface feeder)